Multilevel Inheritance: Multilevel inheritance in Python occurs when a class inherits from a another drived class, creating a chain of inheritance similar to a family tree (Grandparent → Parent → Child).

In this structure, the "grandchild" class has access to all methods and attributes from both its immediate parent and its grandparent (as shown in Figure 3.5).

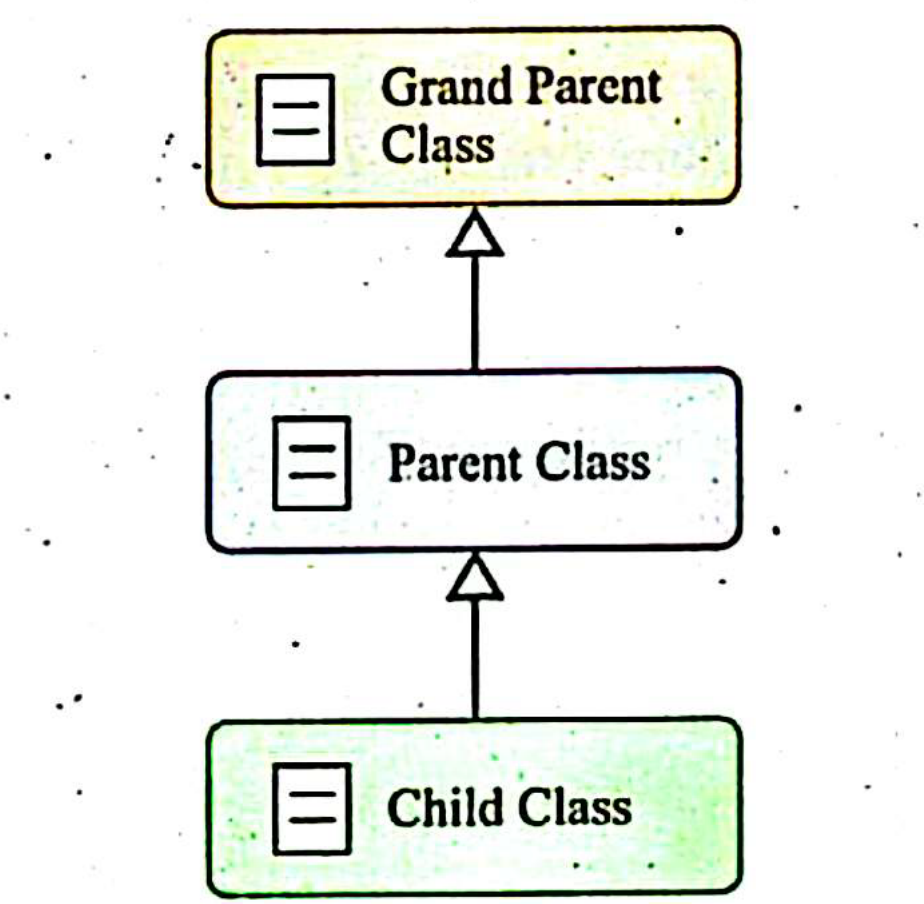


Figure 3.5: Multilevel Inheritance

Example:

In this example, ElectricCar inherits from Car, which in turn inherits from Vehicle.

```
class Vehicle:          # Grandparent Class
    def start_engine(self):
        print("Engine started.")

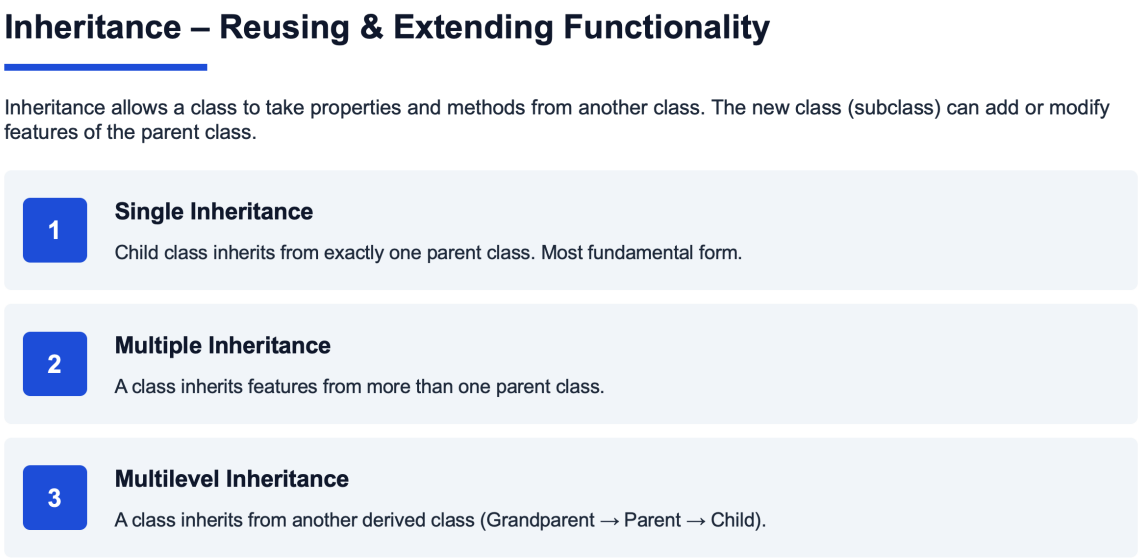
class Car(Vehicle):                 # Parent Class
    def drive(self):
```

```
        print("Car is driving.")

class ElectricCar(Car):                 # Child Class
(Grandchild to Vehicle)
    def charge(self):
        print("Battery is charging.")

my_tesla = ElectricCar()
my_tesla.start_engine()     # Inherited from
Vehicle
my_tesla.drive()            # Inherited from Car
my_tesla.charge()           # Defined in
ElectricCar
```

Method Overriding: A child class can provide a specific implementation of a method that is already provided by its parent. You can use the `super()` function to call the parent's version of that method.



📝 Short Questions

1. Define Multilevel inheritance.
2. Demonstrate Multilevel inheritance through diagram.
3. Distinguish between multiple inheritance and multilevel inheritance.
4. **What three-tier chain illustrates multilevel inheritance in the textbook's code?**
    1. **Answer:** Vehicle (Grandparent) → Car (Parent) → ElectricCar (Child).

🖋️ Long Questions

1. Describe the concept of inheritance and analyze its key variations (Single, Multiple, and Multilevel). Illustrate your answer with diagrams showing how structural inheritance lines connect.
2. **Compare multiple inheritance and multilevel inheritance, explaining their structural differences.**
    1. **Answer:** Multiple and multilevel inheritance structure code relationships differently:
        1. **Multiple Inheritance:** Connects a single child class to two or more independent parent classes at the same time. This allows it to combine distinct sets of behaviors into a hybrid class, like a `Child` class inheriting traits from both `Father` and `Mother`.
        2. **Multilevel Inheritance:** Follows a linear, top-down vertical chain resembling a family tree. A child class inherits from a parent class, which in turn inherits from a grandparent class. This gives the bottom grandchild class access to all attributes and methods across every level of the chain.

Class Activity:

Create a Vehicle class and extend it with Car and Bike subclasses. Add unique methods to each subclass and demonstrate inheritance.